

Basic Concepts

1. Current: Electric current is the rate of flow of electric charge per unit time.

Formula -

$$I = \frac{\Delta q}{\Delta t} \quad (A/C^{31})$$

$$\Rightarrow I = \frac{dq}{dt}$$

$$\Rightarrow dq = I dt$$

$$\Rightarrow q = \int_{t_1}^{t_2} I dt \quad (C)$$

where,

I = Current (Ampere)

q = total charge (Coulomb)

t = time (second)

Δq = change in charge

Δt = time interval
(অবধি সময়)
 $= (t_2 - t_1)$

2. Voltage: Voltage is the work done per unit charge to move a charge between two points.

Formula -

$$V = \frac{\Delta w}{\Delta q} \quad (V/JC^{-1})$$

$$\Rightarrow V = \frac{dw}{dq}$$

$$\Rightarrow dw = V dq$$

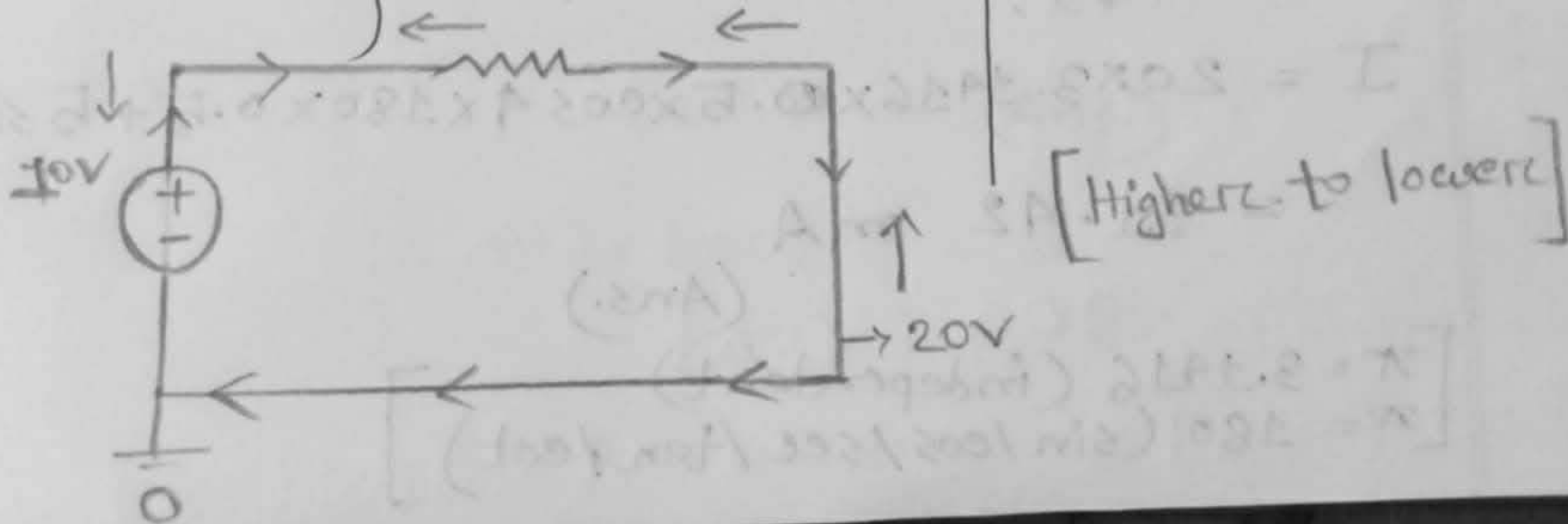
$$\Rightarrow w = \int V dq \quad (J)$$

where,

V = Voltage (volt)

w = work/Energy (Joule)

q = charge (Coulomb)



3. Power: Electric power is the rate at which electrical energy is consumed or work is done.

Formula —

$$P = \frac{\Delta w}{\Delta t} \quad (W / s^1)$$

$$\Rightarrow P = \frac{\Delta q}{\Delta t} \cdot \frac{\Delta w}{\Delta q} \\ = I \cdot V \quad (VA)$$

where,

P = Power (Watt)

w = Energy (Joule)

t = time (sec)

Example —

1.2] Given that,

$$q = 5t \sin 4\pi t \text{ mC}$$

We know,

$$I = \frac{\Delta q}{\Delta t}$$

$$= \frac{d}{dt} (5t \sin 4\pi t)$$

$$= 5t \cdot \cos 4\pi t \cdot 4\pi + \sin 4\pi t \cdot 5$$

$$= 20\pi t \cdot \cos 4\pi t + 5 \sin 4\pi t$$

At $t = 0.5s$,

$$I = 20 \times 3.1416 \times 0.5 \times \cos 4 \times 180 \times 0.5 + 5 \sin 4 \times 180 \times 0.5$$

$$= 31.42 \text{ mA}$$

(Ans.)

$$\left[\begin{array}{l} \pi = 3.1416 \text{ (independent)} \\ \pi = 180 \text{ (sin/cos/sec/tan/cot)} \end{array} \right]$$

1.3 Given that,

$$t_1 = 1s, \quad t_2 = 2s, \quad I = (3t^2 - t) A$$

We know,

$$q = \int_{t_1}^{t_2} I dt$$

$$= \int_1^2 (3t^2 - t) dt$$

$$= \left[3 \cdot \frac{t^3}{3} - \frac{t^2}{2} \right]_1^2$$

$$= \left(2^3 - \frac{2^2}{2} \right) - \left(1^3 - \frac{1^2}{2} \right)$$

$$= 8 - 2 - 1 + \frac{1}{2}$$

$$= 5.5 C$$

(Ans.)

1.4 Given that,

$$\Delta w = 2.3 \times 10^3 J, \quad I = 2A, \quad \Delta t = 10s$$

We know,

$$\text{Voltage drop, } V = \frac{\Delta w}{\Delta q}$$

$$= \frac{2.3 \times 10^3}{20}$$

$$= 115 V$$

$$\Delta q = I \cdot \Delta t$$

$$= 2 \times 10$$

$$= 20 C$$

(Ans.)

1.5] Given that, (a)

$$I = 5 \cos 60\pi t \text{ A}$$

$$t = 3 \text{ ms}$$

$$= 3 \times 10^{-3} \text{ s}$$

$$(a) v = 3I$$

$$(b) v = 3 \frac{dI}{dt}$$

(a) We know,

$$P = VI$$

$$= 3I \cdot I \quad [\because v = 3I]$$

$$= 3I^2$$

$$= 3 \times (5 \cos 60\pi t)^2$$

$$= 3 \times 25 \cos^2 60\pi t$$

$$= 75 \cos^2 60\pi t$$

$$\text{At } t = 3 \times 10^{-3} \text{ s}$$

$$P = 75 \times \cos^2(60 \times 180 \times 3 \times 10^{-3})$$

$$= 53.467 \text{ W} \quad [\text{absorb Power}]$$

(Ans.)

$$(b) v = 3 \frac{dI}{dt}$$

$$= 3 \times \frac{d}{dt} (5 \cos 60\pi t)$$

$$= 3 \times 5 \cdot -\sin 60\pi t \times 60\pi$$

$$= -900\pi \sin 60\pi t$$

$$\therefore P = VI = -900\pi \sin 60\pi t \times 5 \cos 60\pi t$$

$$= -1500 \times 3.1416 \times \sin(60 \times 180 \times 3 \times 10^3) \times \cos(60 \times 180 \times 3 \times 10^3)$$

$$[\because t = 3 \times 10^3]$$

$$= -6395.9 \text{ W} \quad [\text{Supply Power}]$$

(Ans.)

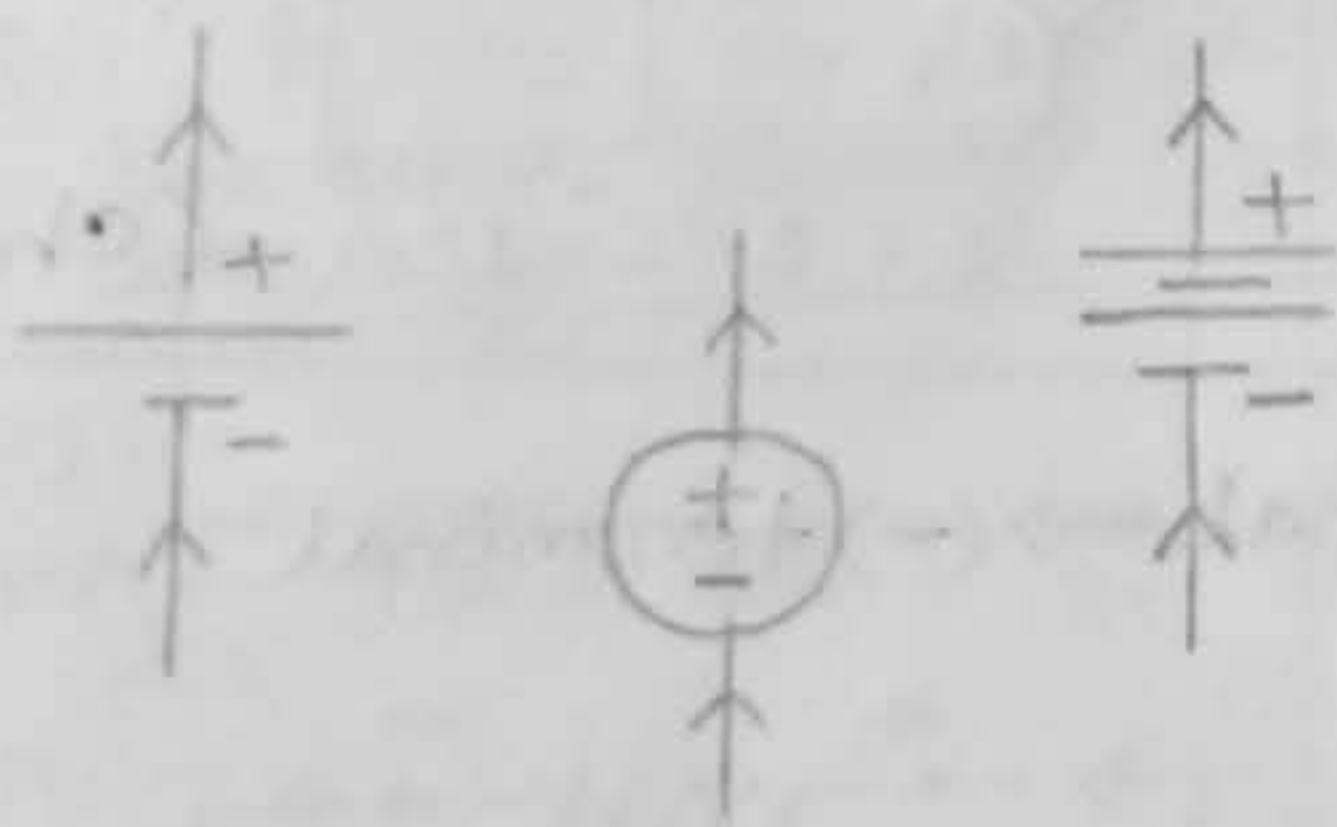
$$* \int 5 \cos 60\pi t = \frac{5 \sin 60\pi t}{60\pi}$$

$$* \frac{d}{dt} (5 \cos 60\pi t) = -5 \sin 60\pi t \cdot 60\pi$$

- DC Source: It is a type of electric current that flows in only one direction and usually has constant voltage. (Direct current source)

(i) Voltage Source

Independent (আনেক সাধিত্বের সূত্র)



(ii) Current Source

Independent (আনেক সাধিত্বের সূত্র)

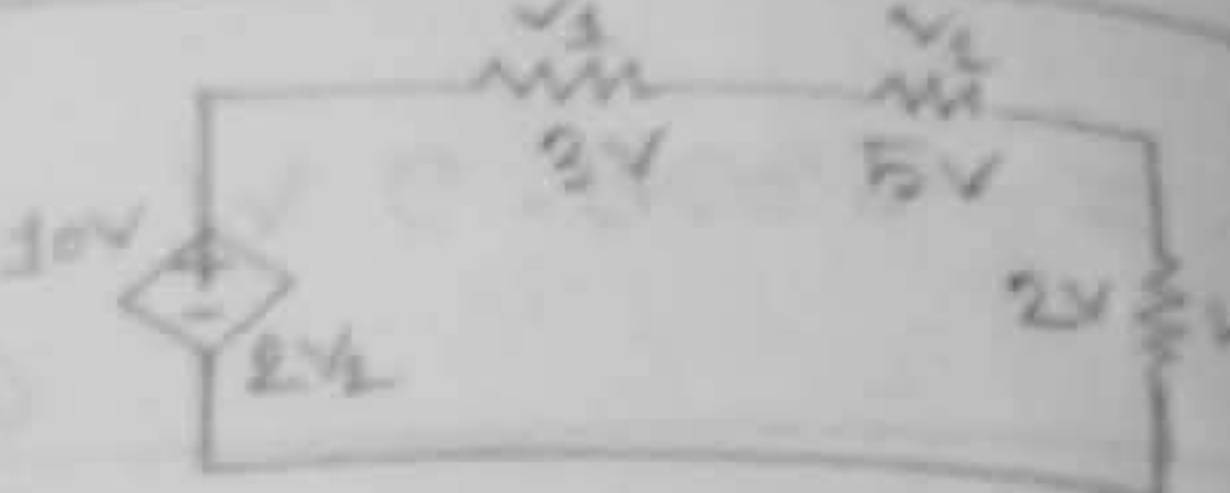
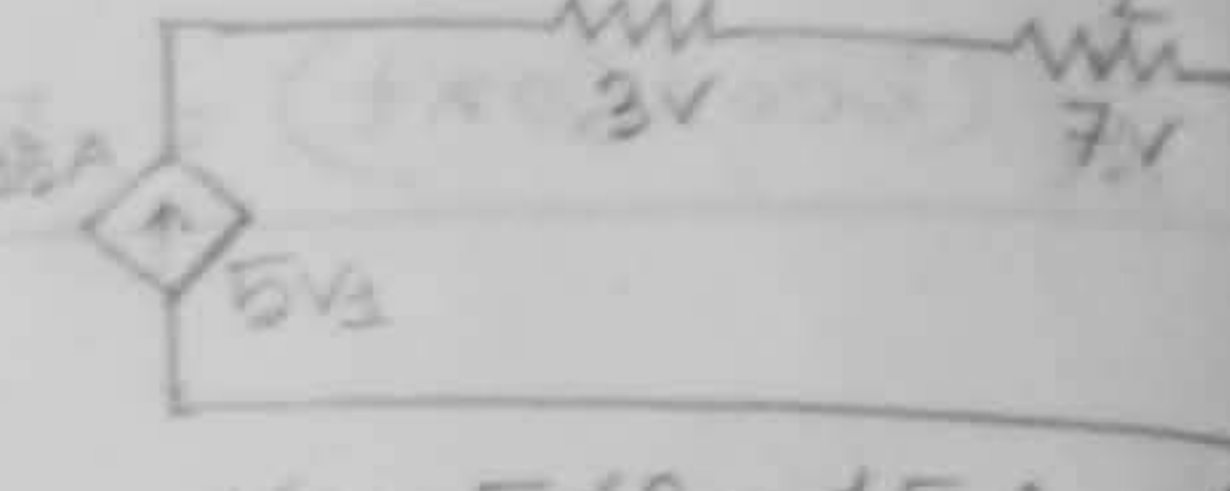
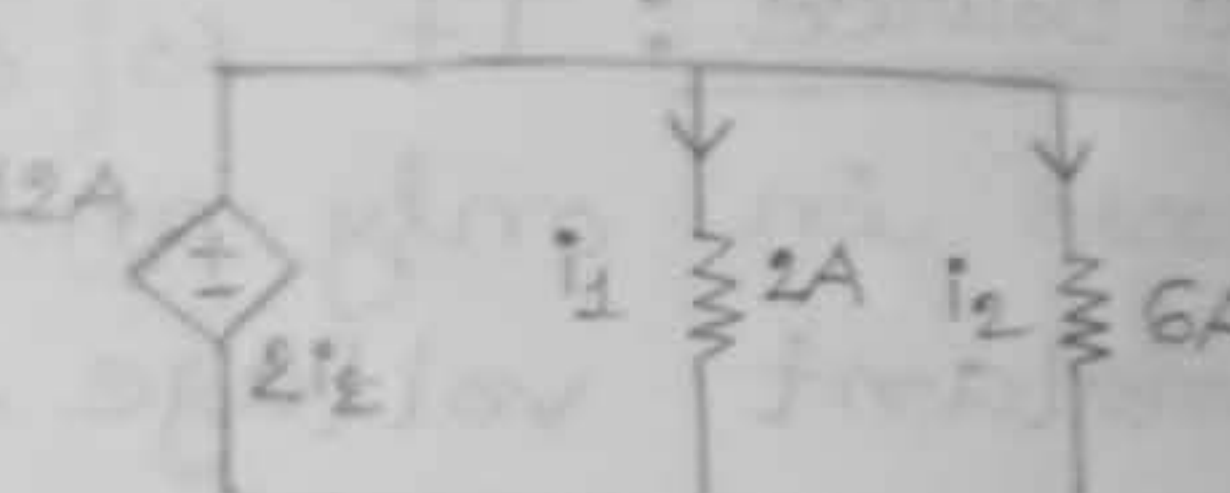
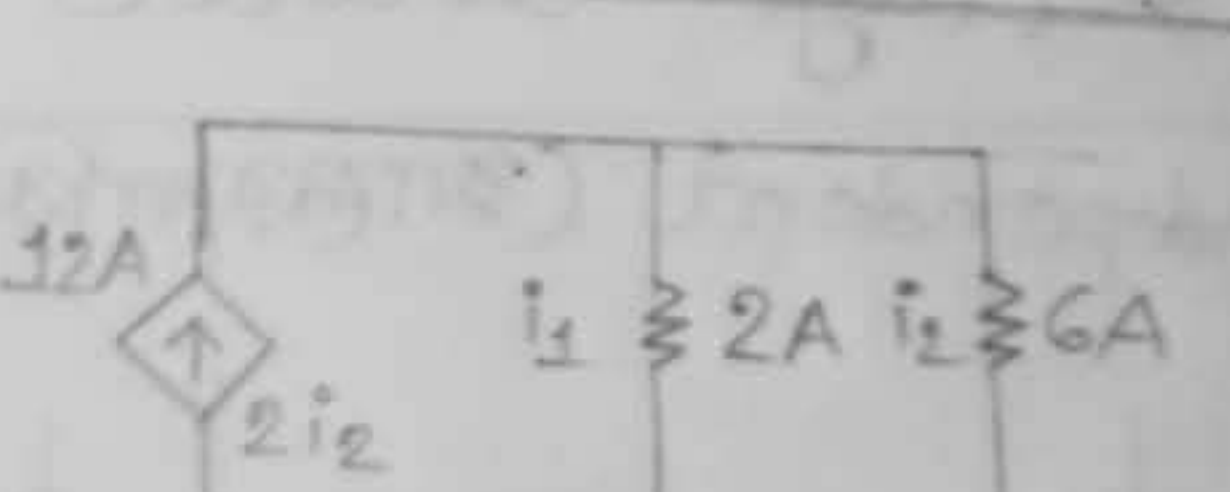


Dependent (আনেক সাধিত্বের সূত্র)

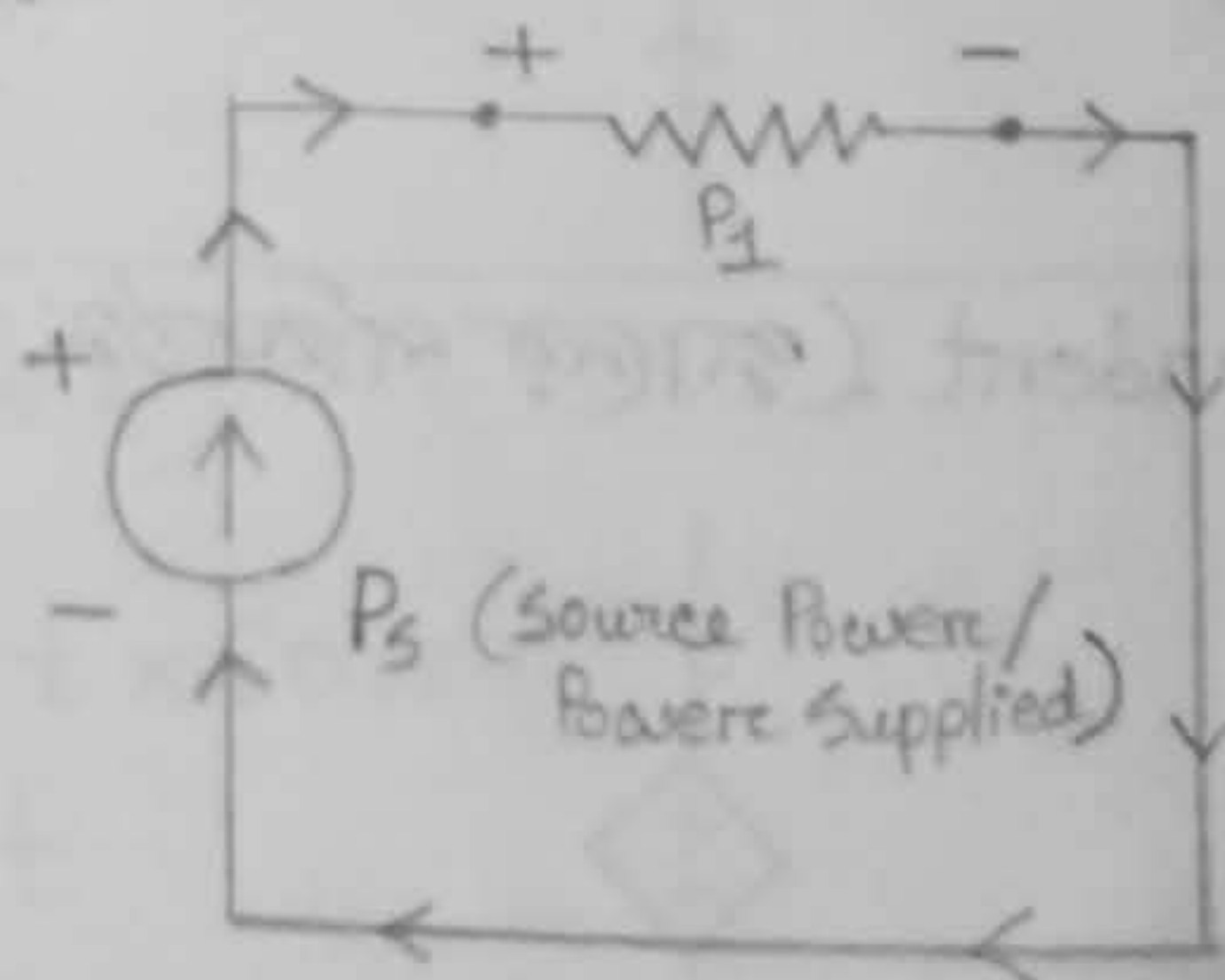
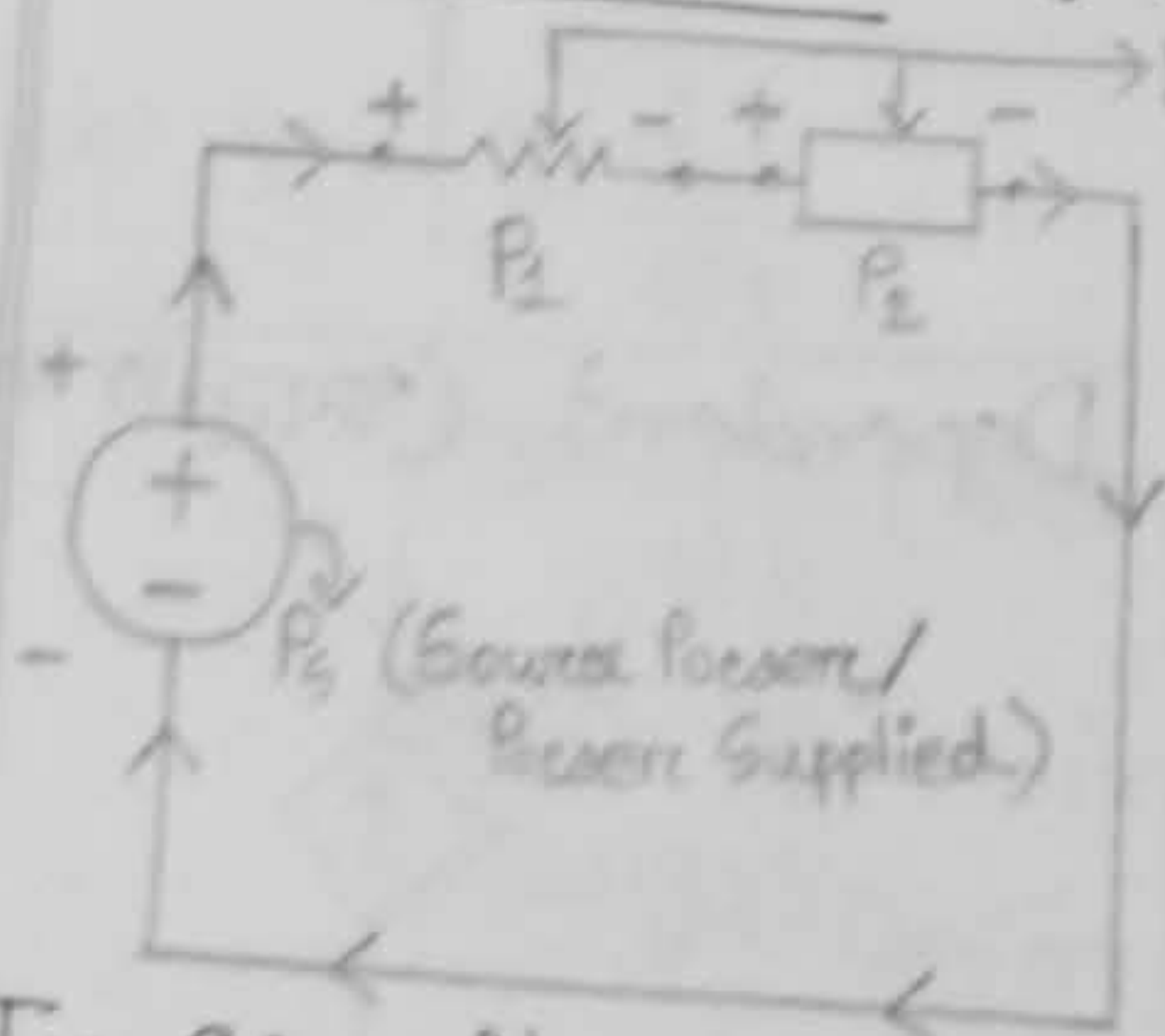


Dependent (আনেক সাধিত্বের সূত্র)



Name	Diagram
1. Voltage Controlled Voltage Source	 $V_2 = 2 \times 5 = 10V$ (Change)
2. Voltage Controlled Current Source	 $V_1 = 5 \times 3 = 15A$ (Change)
3. Current Controlled Voltage Source	 $i_2 = 2 \times 6 = 12A$ (Change)
4. Current Controlled Current Source	 $i_2 = 2 \times 6 = 12A$ (Change)

* Current flows - (+) terminal $\rightarrow$ (-) terminal



In Circuit,

Source Power (P_S) - Absorbed Power ($P_1 + P_2 / P_1$) = 0

or,

$$\boxed{\text{Source Power} = \text{Absorbed Power}}$$

• Passive Sign Convention:

(i) Current (I) $\rightarrow (-) \rightarrow (+) \rightarrow P = -VI \rightarrow \text{Supply Power}$

(ii) Current (I) $\rightarrow (+) \rightarrow (-) \rightarrow P = +VI \rightarrow \text{Absorb Power}$

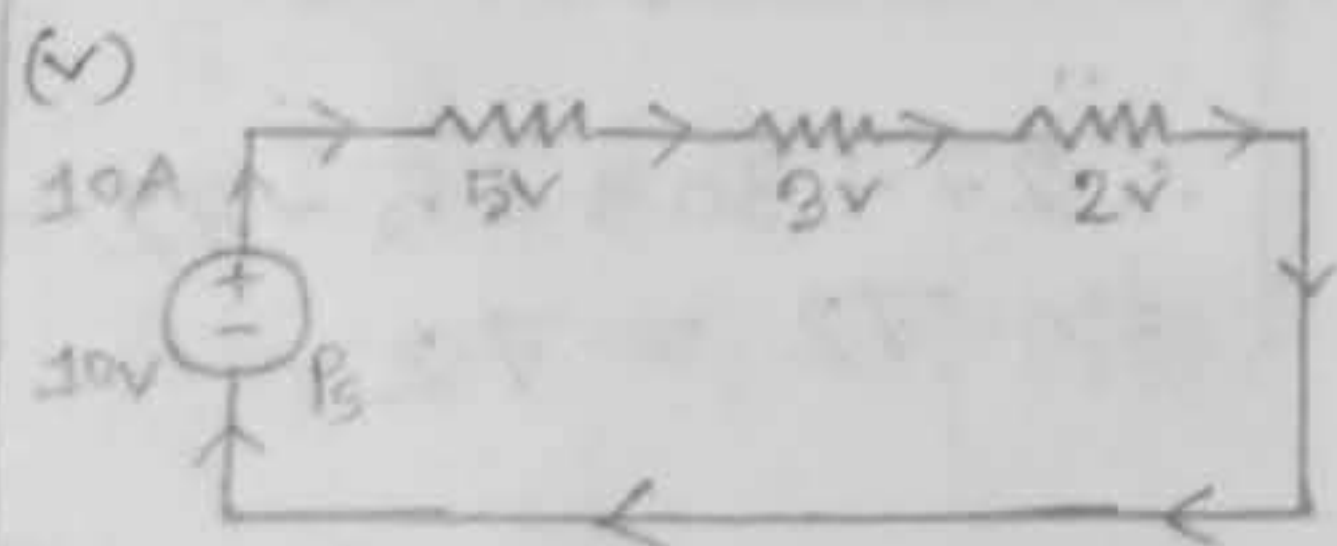
• Series Connection -

(i) Current Same
 $I_1 = I_2 = I_3$

(ii) Voltage Divided
 $V = V_1 + V_2 + V_3$

(iii) $R_T = R_1 + R_2 + R_3$
↳ Total Resistance

(iv) If one resistance fails, the whole circuit stop working.



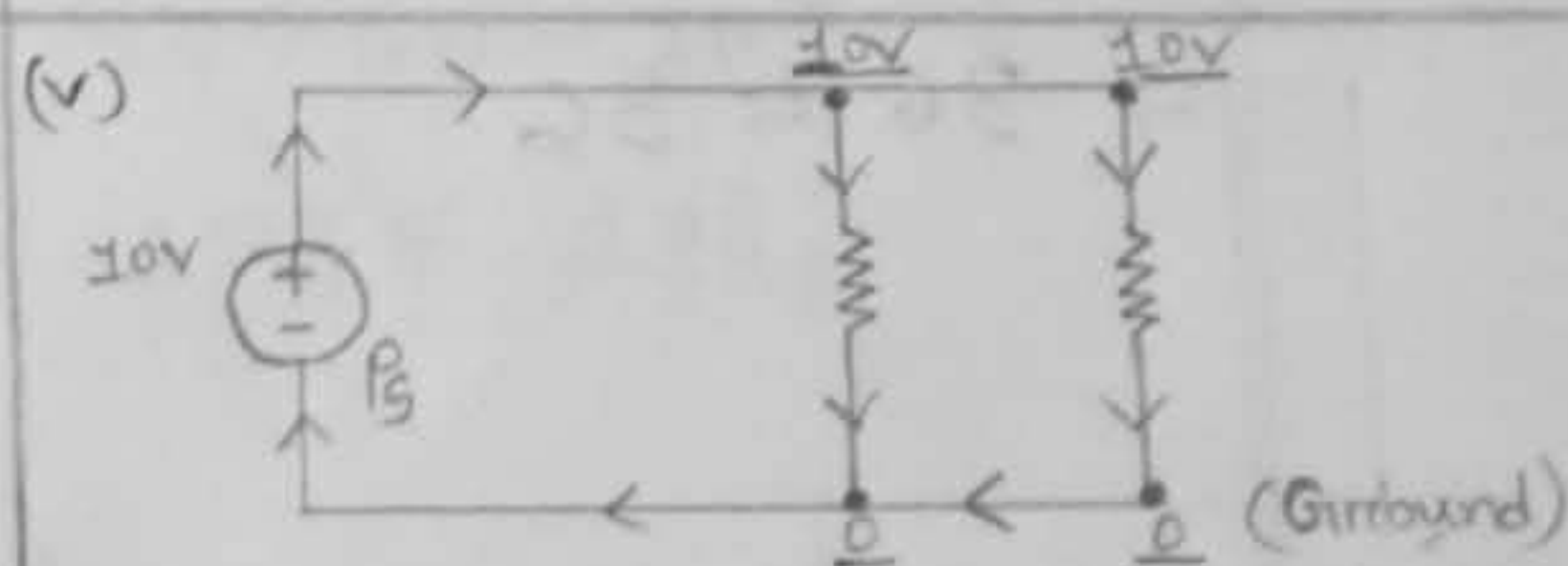
Parallel Connection -

(i) Current divided
 $I = I_1 + I_2 + I_3$

(ii) Voltage Same (যুট্টে সার্কিট
voltage সার্কিট ২(২))

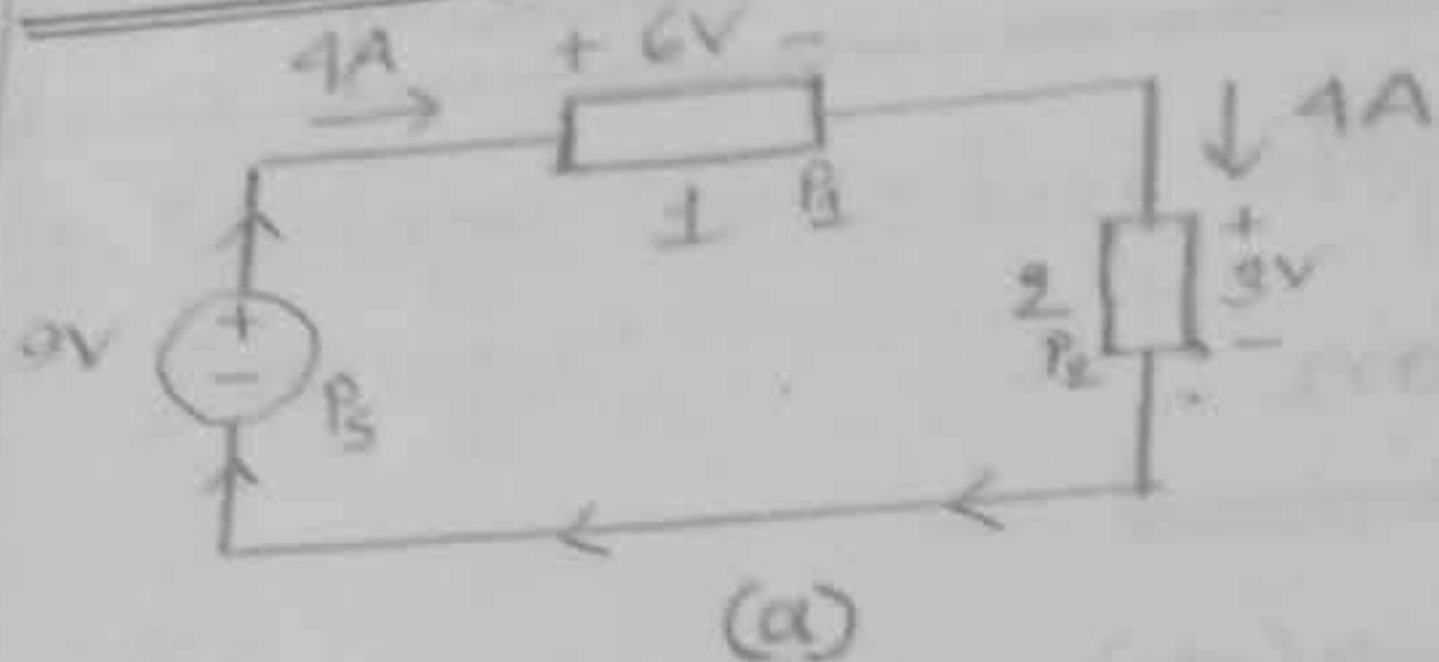
(iii) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

(iv) If one resistance fails, other branches still work.



1.18]

Example —



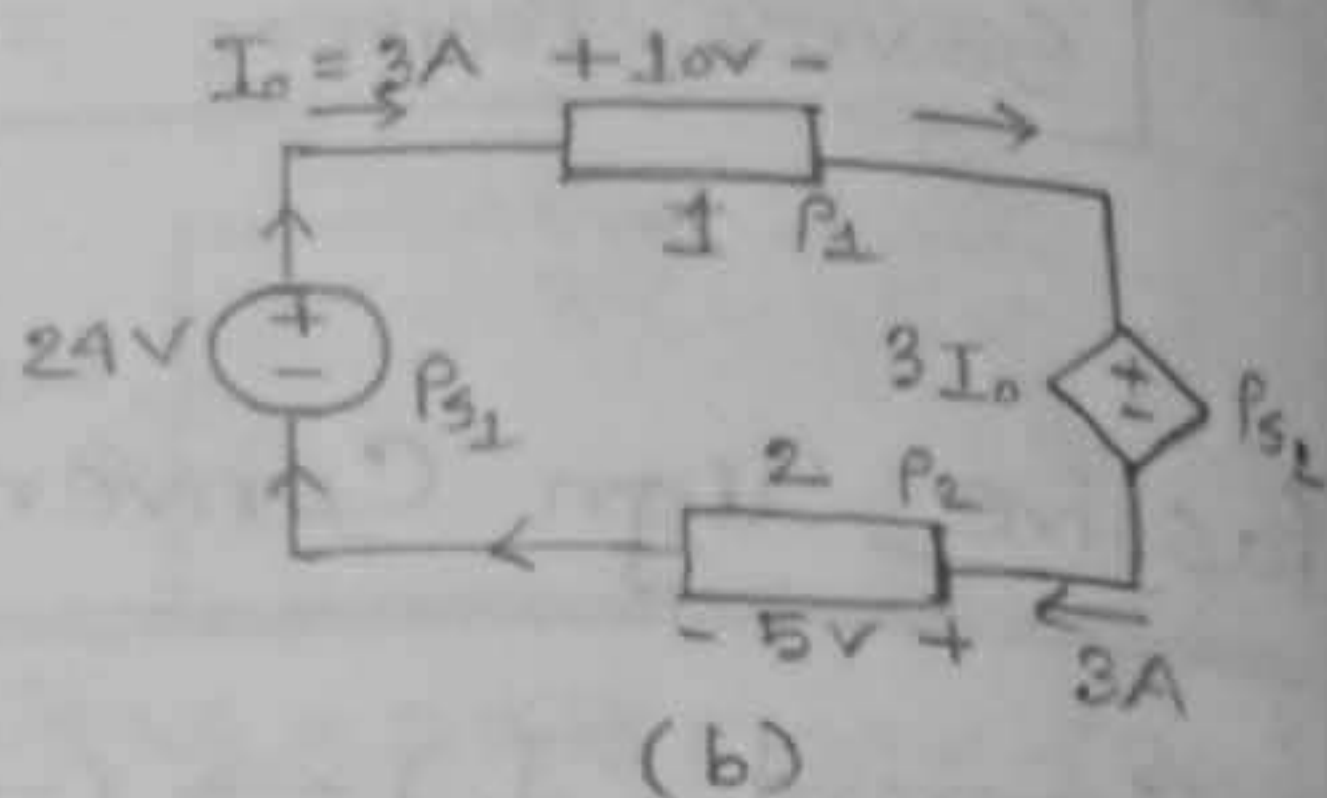
$$\begin{aligned} P_S &= -VI \\ &= -9 \times 4 \\ &= -36 \text{ W (supply)} \end{aligned}$$

$$\begin{aligned} P_1 &= +VI \\ &= 6 \times 4 \\ &= +24 \text{ W (absorb)} \end{aligned}$$

$$\begin{aligned} P_2 &= +VI \\ &= 3 \times 4 \\ &= +12 \text{ W (absorb)} \end{aligned}$$

$$\therefore 24 + 12 - 36 = 0$$

$$\Rightarrow 36 = 36$$



$$\begin{aligned} P_{S1} &= -VI \\ &= -24 \times 3 \\ &= -72 \text{ W (supply)} \end{aligned}$$

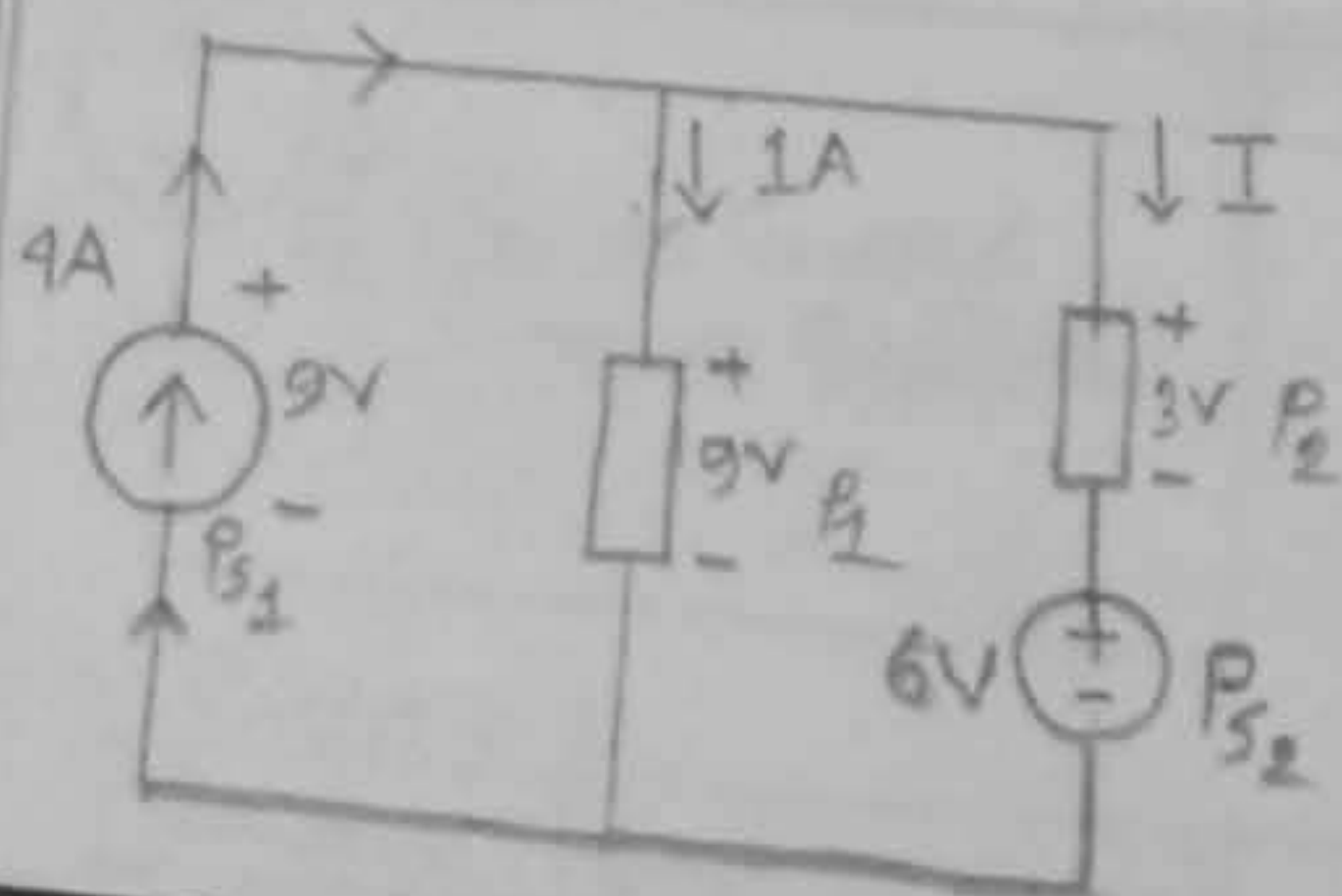
$$\begin{aligned} P_{S2} &= +VI \\ &= +3I_0 \times 3 \\ &= 3 \times 3 \times 3 \\ &= +27 \text{ W (absorb)} \end{aligned}$$

$$\begin{aligned} P_1 &= +VI \\ &= 10 \times 3 \\ &= +30 \text{ W (absorb)} \end{aligned}$$

$$\begin{aligned} P_2 &= +VI \\ &= +5 \times 3 \\ &= +15 \text{ W (absorb)} \end{aligned}$$

$$\begin{aligned} \therefore 27 + 30 + 15 - 72 &= 0 \\ \Rightarrow 72 &= 72 \end{aligned}$$

1.19]



$$\begin{aligned} P_{S1} &= -VI \\ &= -9 \times 4 \\ &= -36 \text{ W (supply)} \end{aligned}$$

$$\begin{aligned} P_{S2} &= +VI \\ &= +6 \times I \end{aligned}$$

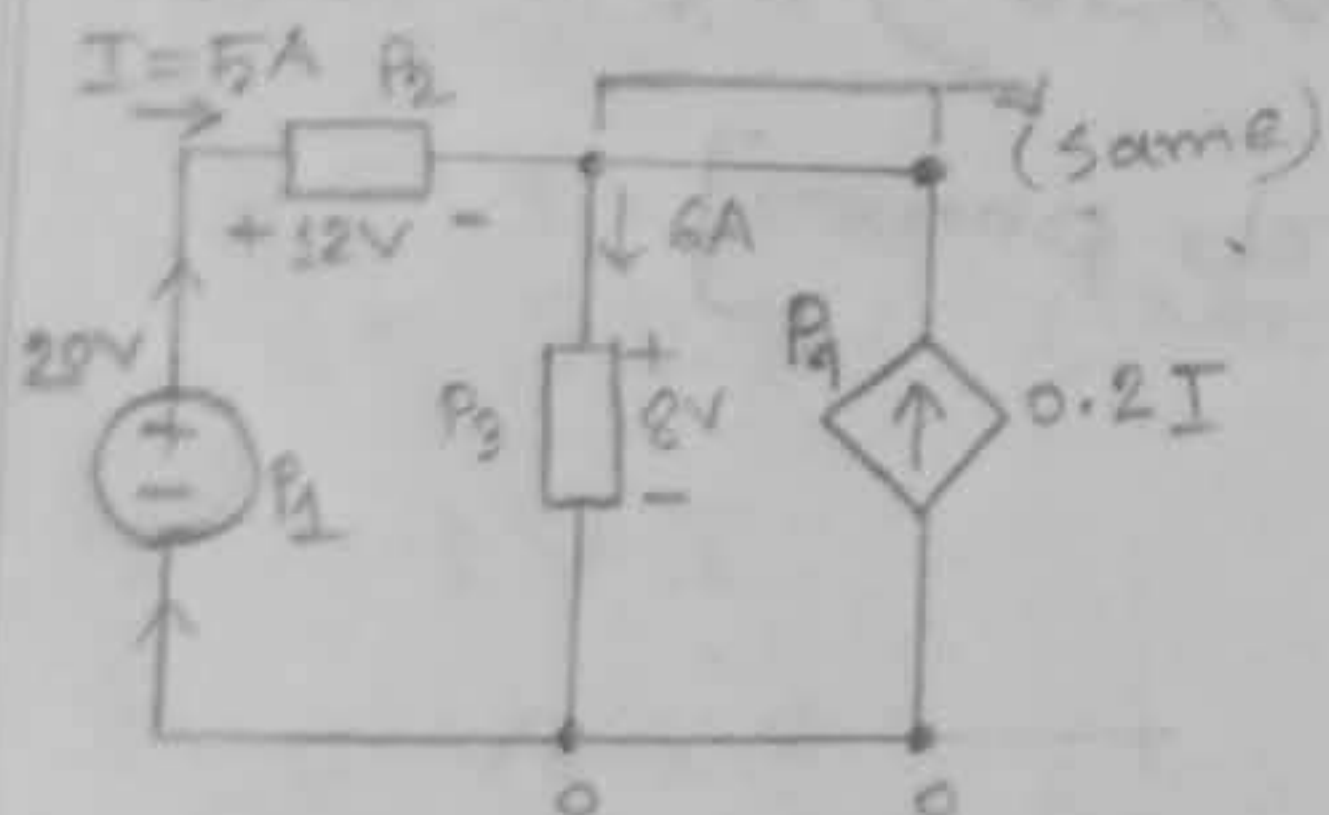
$$= 6I \text{ (absorb)}$$

$$\begin{aligned} P_1 &= +VI \\ &= +9 \times 1 \\ &= 9 \text{ W (absorb)} \end{aligned}$$

$$\begin{aligned} P_2 &= +VI \\ &= +3 \times I \\ &= 3I \text{ (absorb)} \end{aligned}$$

$$\begin{aligned} \therefore 3I + 9 + 6I - 36 &= 0 \\ \Rightarrow 9I - 27 &= 0 \\ \Rightarrow 9I &= 27 \\ \Rightarrow I &= 3 \text{ A} \end{aligned}$$

1.7



$$\begin{aligned} P_1 &= -VI \\ &= -20 \times 5 \\ &= -100 \text{ W (supply)} \end{aligned}$$

$$\begin{aligned} P_2 &= +VI \\ &= 12 \times 5 \\ &= 60 \text{ W (absorb)} \end{aligned}$$

$$\begin{aligned} P_3 &= +VI \\ &= 8 \times 6 \\ &= 48 \text{ W (absorb)} \end{aligned}$$

$$\begin{aligned} P_4 &= -VI \\ &= -0.8I \times 0.2 \times 5 \\ &= -8 \text{ W (supply)} \end{aligned}$$

$$\begin{aligned} \therefore 48 + 60 - 100 - 8 &= 0 \\ \Rightarrow 108 &= 108 \end{aligned}$$

1.5]

Given,

(b)

$$i = 5 \cos 60\pi t \text{ A}$$

$$t = 5 \text{ ms}$$

$$= 5 \times 10^{-3} \text{ s}$$

$$(a) v = 2i$$

$$(b) v = \left(10 + 5 \int_0^t i dt \right) v$$

(a) We know,

$$P = VI$$

$$= 2 \times 5 \cos 60\pi t \times 5 \cos 60\pi t$$

$$\text{at } t = 5 \times 10^{-3} \text{ s,}$$

$$\therefore P = 2 \times 5 \times \cos(60 \times 180 \times 5 \times 10^{-3}) \times 5 \times \cos(60 \times 180 \times 5 \times 10^{-3})$$

$$= 17.27 \text{ W} \quad [\text{absorb power}]$$

(b) We know,

$$P = VI$$

$$= \left(10 + 5 \int_0^t i dt \right) \times I$$

$$= \left[10 + 5 \int_0^t 5 \cos 60\pi t dt \right] \times 5 \cos 60\pi t$$

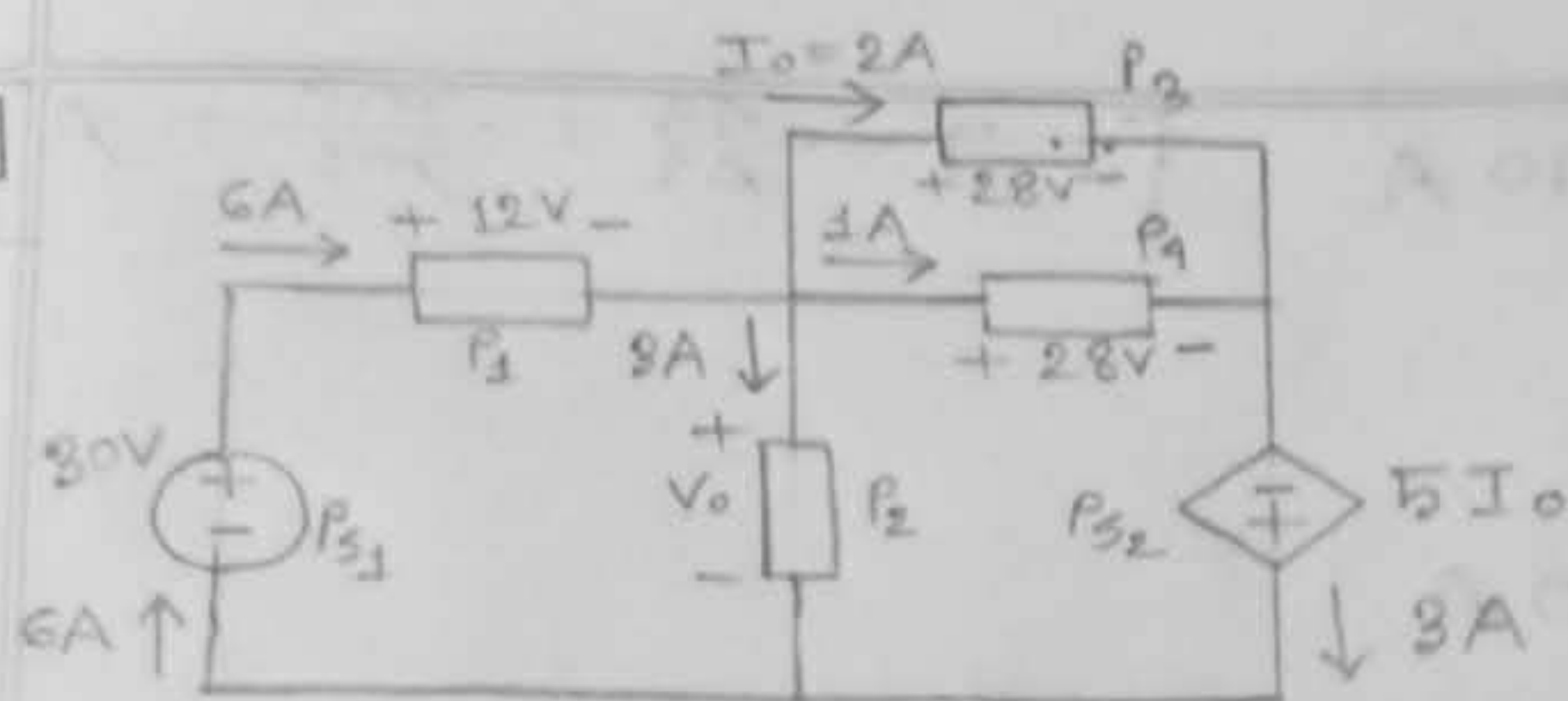
$$= \left[10 + 5 \times \frac{5 \sin 60\pi t}{60\pi} \right] \times 5 \cos 60\pi t$$

$$\text{at } t = 5 \times 10^{-3} \text{ s,}$$

$$\therefore P = \left[10 + 5 \times \frac{5 \sin(60 \times 180 \times 5 \times 10^{-3})}{60 \times 3.1416} \right] \times 5 \cos(60 \times 180 \times 5 \times 10^{-3})$$

$$= 29.7 \text{ W} \quad [\text{absorb power}] \quad (\text{Ans.})$$

1.20



$$P_{51} = -VI$$

$$= -30 \times 6$$

$$= -180 \text{ W (supply)}$$

$$P_{52} = -VI$$

$$= -5 \times 2 \times 3$$

$$= -30 \text{ W (supply)}$$

$$P_1 = +VI$$

$$= 12 \times 6$$

$$= 72 \text{ W (absorb)}$$

$$P_2 = +VI$$

$$= V_0 \times 3$$

$$= 3V_0 \text{ (absorb)}$$

$$P_3 = +VI$$

$$= 28 \times 2$$

$$= 56 \text{ W (absorb)}$$

$$P_4 = +VI$$

$$= 28 \times 1$$

$$= 28 \text{ W (absorb)}$$

$$\therefore 72 + 3V_0 + 56 + 28 = 180 - 30 = 0$$

$$\Rightarrow 156 + 3V_0 = 210$$

$$\Rightarrow 3V_0 = 210 - 156$$

$$\Rightarrow 3V_0 = 54$$

$$\Rightarrow V_0 = \frac{54}{3}$$

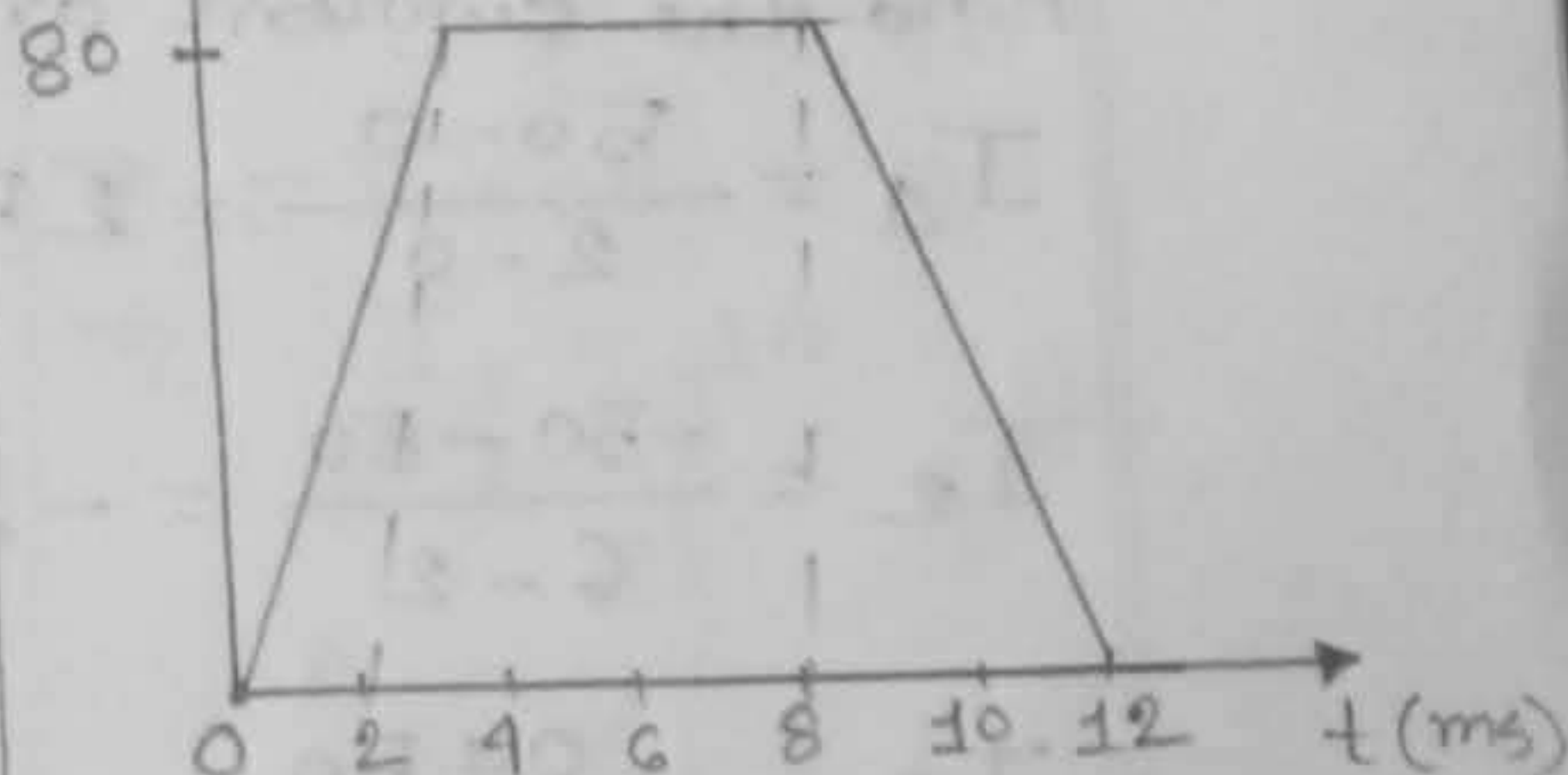
$$\Rightarrow V_0 = 18 \text{ V}$$

(Ans.)

— X —

1.61

q(t) (mC)



Find the current at: -

(a) $t = 1 \text{ ms}$

$$I = \frac{dq}{dt} = \frac{\Delta q}{\Delta t} = \frac{q_2 - q_1}{t_2 - t_1}$$

$$I = \frac{80-0}{2-0} = 40 \text{ A}$$

$$[\because I = \frac{\Delta q}{\Delta t} = \frac{mc}{ms} = A]$$

(b) at $t = 6 \text{ ms}$,

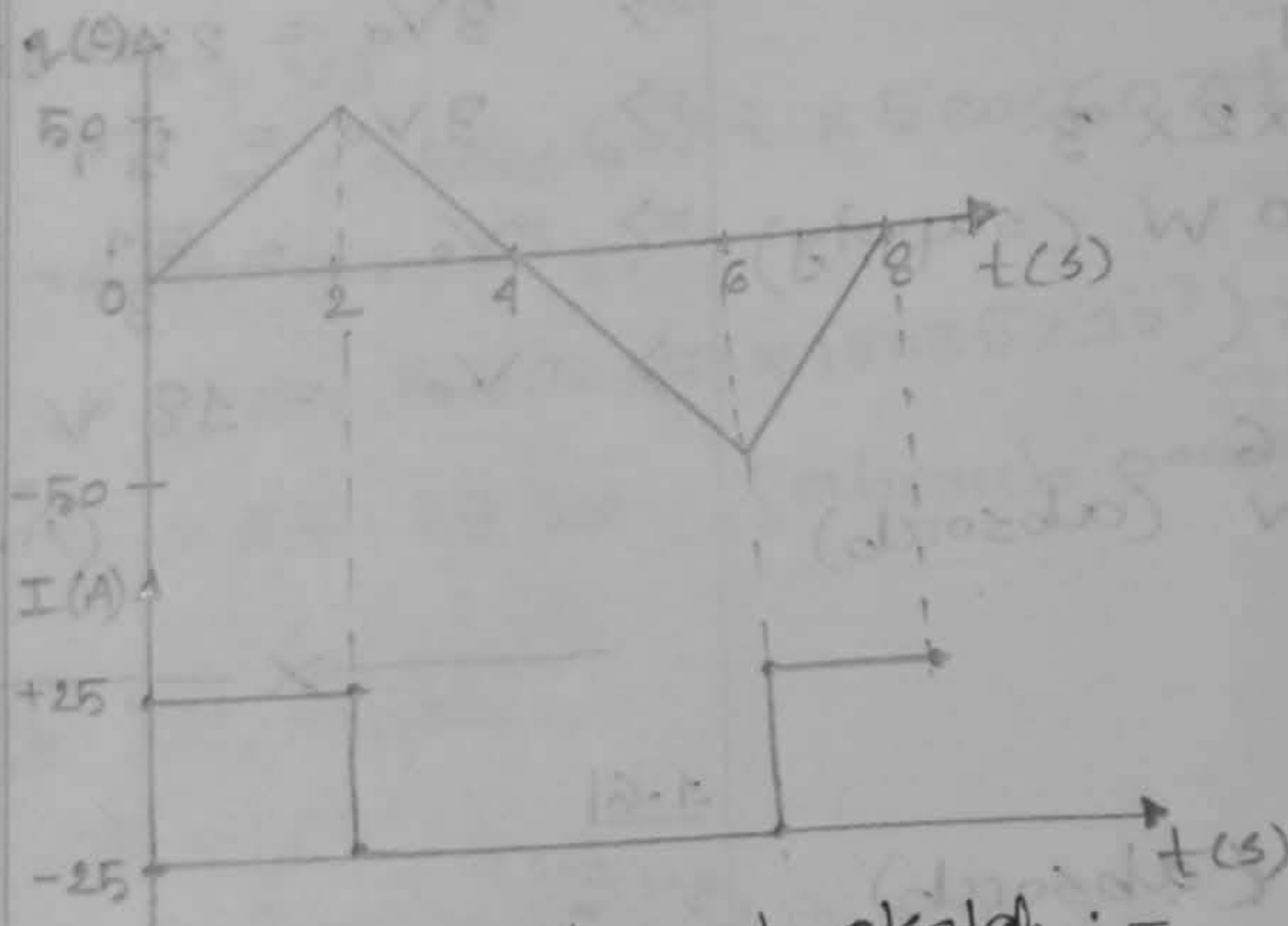
$$I = \frac{80-80}{8-2} = 0 \text{ A}$$

(c) at $t = 10 \text{ ms}$,

$$I = \frac{0-80}{12-8} = -20 \text{ A}$$

(Ans.)

1.7



Find the current and sketch :-

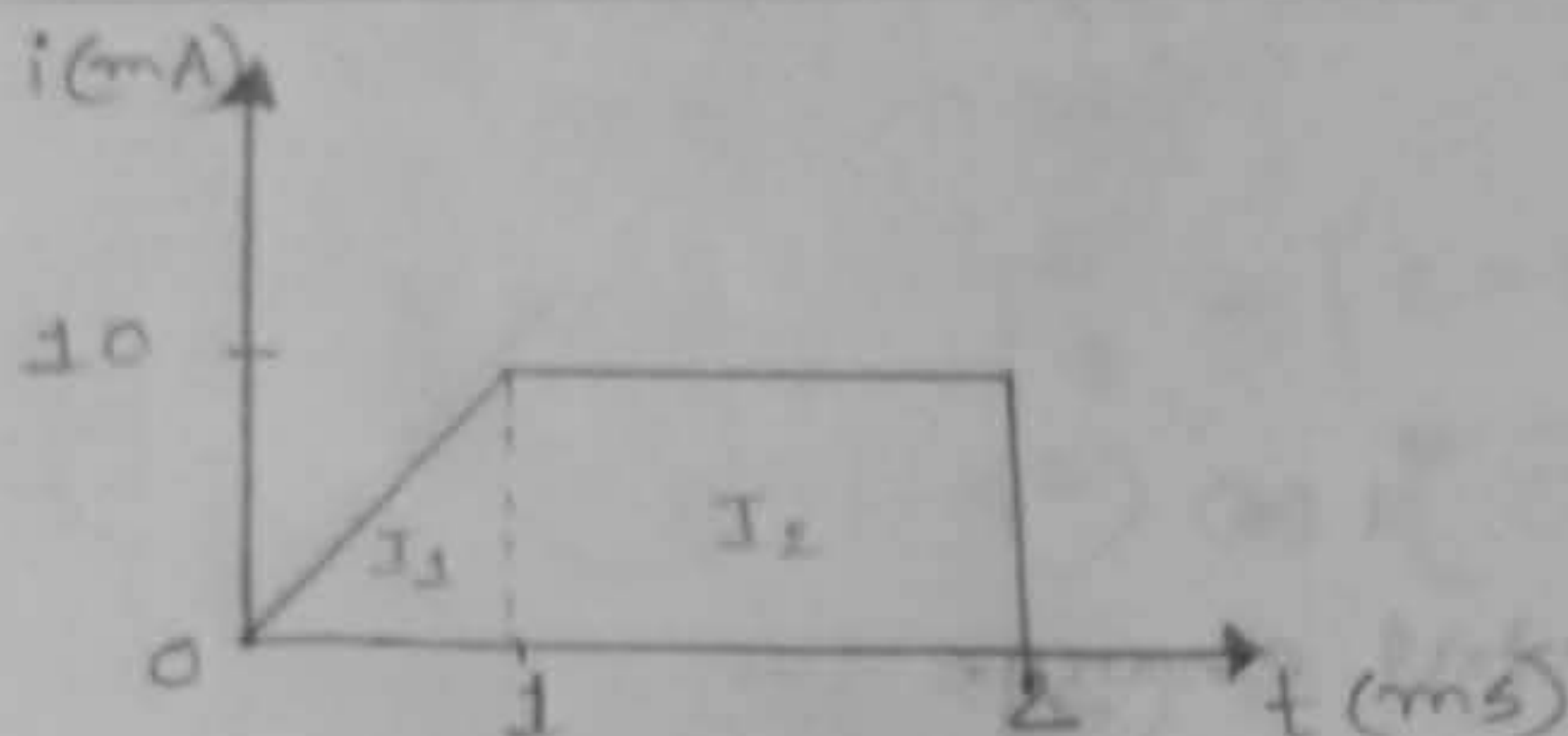
$$I_1 = \frac{50-0}{2-0} = 25 \text{ A}$$

$$I_2 = \frac{-50-50}{6-2} = -25 \text{ A}$$

$$I_3 = \frac{0+50}{8-6} = 25 \text{ A}$$

(Ans.)

1.8



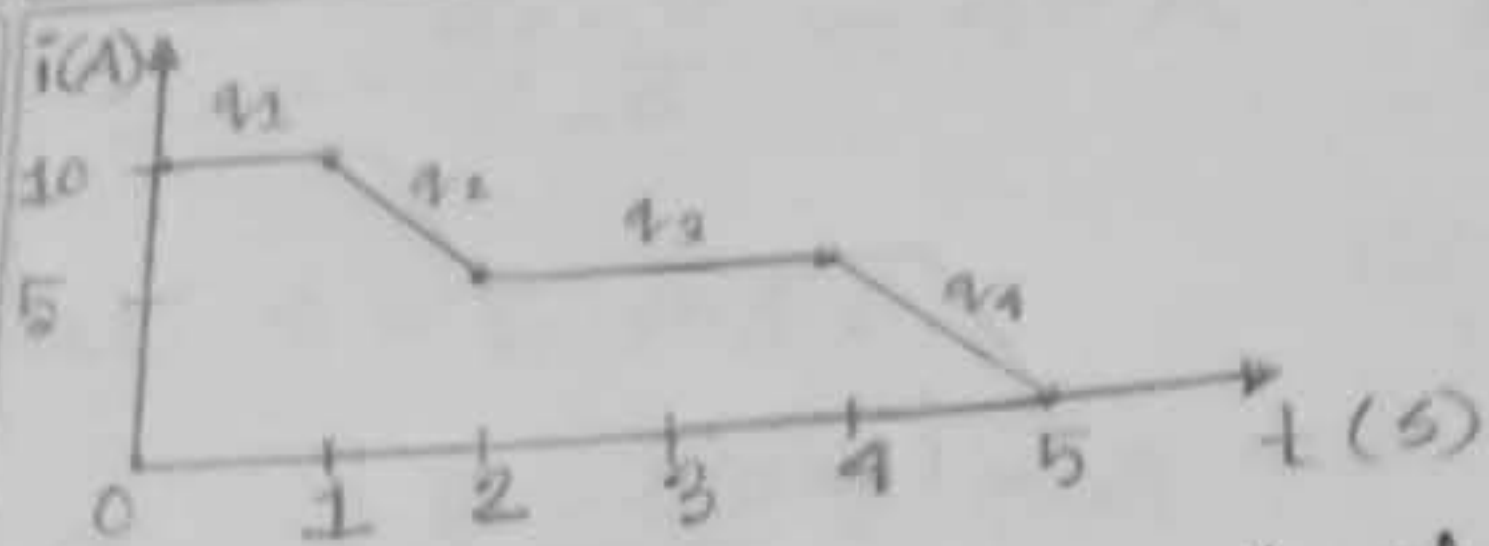
Calculate the total charge:-

$$\begin{aligned}
 q &= \int_{t_2}^{t_1} I dt \\
 &= \int_{t_1}^{t_2} I_1 dt + \int_{t_2}^{t_3} I_2 dt \\
 &= \int_0^1 10t dt + \int_1^2 10 dt \\
 &= \left[10 \times \frac{t^2}{2} \right]_0^1 + \left[10t \right]_1^2 \\
 &= \left(\frac{1}{2} \times 1^2 - \frac{1}{2} \times 0^2 \right) + (10 \times 2 - 10 \times 1) \\
 &= \frac{1}{2} + 10 \\
 &= 10.5 \text{ } \mu\text{C}
 \end{aligned}$$

$$\begin{aligned}
 \left[\because q = It = \text{mA} \times \text{ms} \right. \\
 &= 10^{-3} \times 10^{-3} \\
 &= 10^{-6} \text{ (micro)}
 \end{aligned}$$

$$\begin{aligned}
 \frac{t - t_1}{t_1 - t_2} &= \frac{I - I_1}{I_1 - I_2} \\
 I_1 \Rightarrow \frac{t - 0}{0 - 1} &= \frac{I - 0}{0 - 10} \\
 \Rightarrow \frac{t}{-1} &= \frac{I}{-10} \\
 \Rightarrow -I &= -10t \\
 \Rightarrow I_1 &= 10t \\
 I_2 \Rightarrow \frac{t - t_1}{t_1 - t_2} &= \frac{I - I_1}{I_1 - I_2} \\
 \Rightarrow \frac{t - 1}{1 - 2} &= \frac{I - 10}{10 - 10} \\
 \Rightarrow \frac{t - 1}{-1} &= \frac{I - 10}{0} \\
 \Rightarrow -I + 10 &= 0 \\
 \Rightarrow I_2 &= 10
 \end{aligned}$$

1.9]



Determine the total charge :-

$$q_1 = \int_{t_1}^{t_2} I dt$$

$$= \int_0^1 10 dt$$

$$= [10t]_0^1$$

$$= (10 - 0)$$

$$= 10 \text{ C}$$

$$q_2 = \int_{t_2}^{t_3} 5t dt$$

$$= \int_1^2 5t dt$$

$$= 5 \left[\frac{t^2}{2} \right]_1^2$$

$$= \frac{5}{2} (4 - 1)$$

$$= \frac{15}{2}$$

$$= 7.5 \text{ C}$$

$$q_3 = \int_2^4 5 dt$$

$$= [5t]_2^4$$

$$= (5 \times 4 - 5 \times 2)$$

$$= 20 - 10$$

$$= 10 \text{ C}$$

$$\frac{t - t_1}{t_1 - t_2} = \frac{I - I_1}{I_1 - I_2}$$

$$(0 - 1s) \Rightarrow \frac{t - 0}{0 - 1} = \frac{I - 10}{10 - 10}$$

$$\Rightarrow \frac{t}{-1} = \frac{I - 10}{0}$$

$$\Rightarrow -I + 10 = 0$$

$$\Rightarrow I = 10 \text{ A}$$

$$(1 - 2s)$$

$$\Rightarrow \frac{t - 1}{1 - 2} = \frac{I - 5}{5 - 10}$$

$$\Rightarrow \frac{t - 1}{-1} = \frac{I - 5}{-5}$$

$$\Rightarrow -5t + 5 = -I + 5$$

$$\Rightarrow I = 5t \text{ A}$$

$$(2 - 4s)$$

$$\Rightarrow \frac{t - 2}{2 - 4} = \frac{I - 5}{5 - 5}$$

$$\Rightarrow \frac{t - 2}{-2} = \frac{I - 5}{0}$$

$$\Rightarrow -2I + 10 = 0$$

$$\Rightarrow I = 5 \text{ A}$$

$$(4 - 5s)$$

$$\Rightarrow \frac{t - 4}{4 - 5} = \frac{I - 0}{0 - 5}$$

$$\Rightarrow -5t + 20 = -I$$

$$\Rightarrow I = 5t - 20$$

29.1.26

$$\begin{aligned} q_1 &= \int_4^5 (5t - 20) dt \\ &= \left[5 \times \frac{t^2}{2} \right]_4^5 - [20t]_4^5 \\ &= \frac{5}{2} (5^2 - 4^2) - (20 \times 5 - 20 \times 4) \\ &= \frac{5}{2} (25 - 16) - (100 - 80) \\ &= 22.5 - 20 \\ &= 2.5 \text{ C} \end{aligned}$$

(a) at $t = 1\text{s}$,

$$q = 10 \text{ C}$$

(b) at $t = 3\text{s}$,

$$\begin{aligned} q &= 10 + 7.5 + 5 \\ &= 22.5 \text{ C} \end{aligned}$$

(c) at $t = 5\text{s}$,

$$\begin{aligned} q &= 10 + 7.5 + 10 + 2.5 \\ &= 30 \text{ C} \end{aligned}$$

(Ans.)